

Evaluate the following (Definite integral).

$$\textcircled{1} \int_0^1 \frac{4}{t^2+1} dt$$

$$= 4 \arctan t \Big|_0^1$$

$$= 4 [\arctan 1 - \arctan 0]$$

$$= 4 \left(\frac{\pi}{4} \right) - 4(0) = \pi$$

$$\textcircled{2} \int_{-1}^1 e^{u+1} du$$

$$= \int_{-1}^1 e^1 \cdot e^u du$$

$$= e e^u \Big|_{-1}^1$$

$$= e^2 - 1$$

Effectiveness of FTC depends on having a good supply of antiderivatives

The indefinite integral:

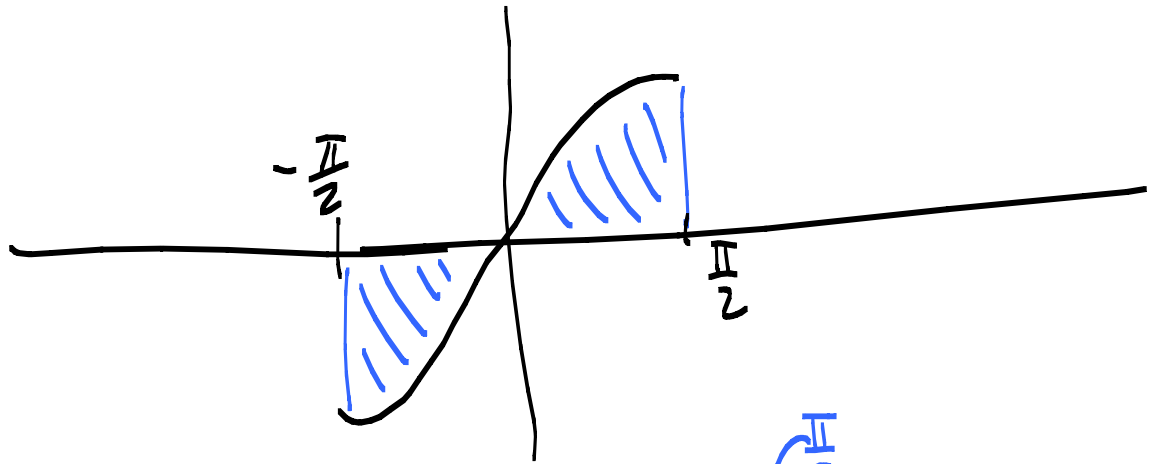
$$\int f(x) dx = F(x) + C$$

Three types of questions:

$$1. \int \sin x dx = -\cos x + C$$

$$2. \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin x dx = -\cos x \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \\ = -\cos\left(\frac{\pi}{2}\right) + \cos\left(-\frac{\pi}{2}\right) = 0$$

3. Find the area between the graph and the x-axis of $f(x) = \sin x$ and $x = -\pi/2$ and $\pi/2$



$$\begin{aligned} \text{Area} &= -\int_{-\pi/2}^0 \sin x \, dx + \int_0^{\pi/2} \sin x \, dx \\ &= -(-1) + 1 = 2 \end{aligned}$$

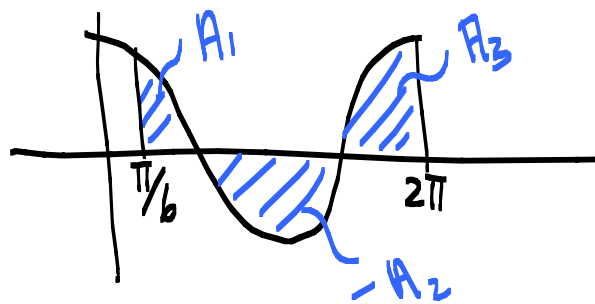
Determine:

1. $\int \cos x \, dx = \sin x + \underline{\underline{C}}$

2. $\int_{\pi/6}^{2\pi} \cos x \, dx = \left(-\frac{1}{2}\right) \leadsto \text{Number.}$

3. The area between the curve of $y = \cos x$ the x-axis and the lines $x = \pi/6$ and $x = 2\pi$

$$\text{Area} = 3\frac{1}{2}.$$



Find the following integrals:

$$1. \int x^5 dx$$
$$= \frac{1}{6} x^6 + C$$

$$2. -\frac{1}{2} \int -2 (1-2x)^5 dx$$
$$= -\frac{1}{2} \left[\frac{1}{6} (1-2x)^6 \right] + C$$

$$3. \int \sin^5 2x \cdot \cos 2x dx$$
$$= \frac{1}{2} \int (\sin 2x)^5 \cdot 2 \cos 2x dx$$
$$= \frac{1}{2} \left[\frac{1}{6} (\sin 2x)^6 \right] + C$$

$$1. \int [f(x)]^n f'(x) dx = \frac{[f(x)]^{n+1}}{n+1} + C$$

$$1. \int \frac{1}{\sqrt[3]{1-2x}} dx$$
$$= -\frac{1}{2} \int -2 (1-2x)^{-1/3} dx$$
$$= -\frac{1}{2} \left[\frac{3}{2} (1-2x)^{2/3} \right] + C$$
$$= -\frac{3}{4} (1-2x)^{2/3} + C$$

$$2. \int [\sin x]^1 \cos x \, dx.$$

$$= \frac{1}{2} (\sin x)^2 + C$$

$$\frac{1}{4} \int 4x^3 e^{x^4} \, dx$$

$$= \frac{1}{4} e^{x^4} + C$$

$$\frac{d}{dx} e^{x^4} = e^{x^4} \cdot 4x^3$$

$$2. \int e^{f(x)} \cdot f'(x) \, dx = e^{f(x)} + C$$

$$1. -\frac{1}{2} \int e^{\cos 2x} \cdot -2 \sin 2x \, dx$$

$$= -\frac{1}{2} [e^{\cos 2x}] + C$$

Test!!

$$2. \int e^{\tan x} \sec^2 x \, dx$$

$$= e^{\tan x} + C$$

$$3. \frac{1}{2} \int \left[\tan 2x \right]^1 \cdot 2 \sec^2 2x \, dx$$

$$\frac{1}{2} \int \tan^2 2x \cdot \sec^2 2x \, dx$$

$$= \frac{1}{2} \cdot \frac{1}{3} \tan^3 2x + C$$

$$= \frac{1}{2} \cdot \frac{1}{2} [\tan 2x]^2 + c \quad \text{Check!!}$$

$$\int \tan^2 2x \sec^3 2x \, dx ? \quad \text{WTW 164.}$$

$$\int \sin \underline{f(x)} f'(x) \, dx = -\cos f(x) + c.$$

$$\int \sin x \, dx = -\cos x + c$$

$$\int \sec^2(\underline{e^x}) \cdot \underline{e^x} \, dx = \tan(e^x) + c$$

$$\int \frac{\sin(\ln x)}{x} \, dx = \int \sin(\underline{\ln x}) \cdot \frac{1}{x} \, dx = -\cos(\ln x) + c$$

$$\int \frac{1}{x^2} \cos\left(\frac{1}{x}\right) \, dx = -\sin\left(\frac{1}{x}\right) + c$$

$$\frac{d}{dx} x^{-1} = -1 x^{-2} = -\frac{1}{x^2}$$

$$\int \frac{\sin 3x}{\sqrt{1 - \cos 3x}} \, dx = \frac{1}{3} \int 3 \sin 3x \left(1 - \cos 3x\right)^{-1/2} \, dx$$

$$\begin{aligned}
 & \int \frac{\sin 3x}{\sqrt{1-\cos^2 3x}} dx = \frac{1}{3} \left[(1-\cos 3x)^{1/2} \cdot 2 \right] + C \\
 & = \frac{2}{3} (1-\cos 3x)^{1/2} + C \\
 & = \int \frac{\sin 3x}{\underbrace{(1-\cos^2 3x)^{-1/2}}_{2\cos 3x}} dx \quad \checkmark \sin 3x \cdot 3
 \end{aligned}$$

$$3. \int \sin f(x) \cdot f'(x) dx = -\cos f(x) + C$$

$$1. \int \frac{1}{x} dx = \ln|x| + C$$

$$2. \int \frac{-1}{1-2x} dx = -\frac{1}{2} \ln|1-2x| + C$$

$$3. \int \frac{3 \sin 3x}{1-\cos 3x} dx = \frac{1}{3} \ln|1-\cos 3x| + C$$

$$\begin{aligned}
 4. \int \frac{1}{1-2x} dx &= -\frac{1}{2} \int (1-2x)^{-1} dx \\
 &= \frac{(1-2x)^0}{0} \quad |||||
 \end{aligned}$$

$$4. \int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C$$

$$1. \frac{1}{2} \int \frac{2x}{1+x^2} dx$$

$$= \frac{1}{2} \ln(1+x^2) + C$$

$$2. 3 \int \frac{1}{1+x^2} dx$$

$$= 3 \arctan x + C$$

$$3. \int \frac{x}{1+x^4} dx = \frac{1}{2} \int \frac{2x}{1+(x^2)^2} dx$$

$$= \frac{1}{2} \arctan(x^2) + C$$

$$5. \int \frac{f'(x)}{1+[f(x)]^2} dx = \arctan f(x) + C$$

$$1. -\frac{1}{7} \int \frac{1}{\sqrt{1-7x}} dx$$

$$= -\frac{1}{7} \left[2(1-7x)^{1/2} \right] + C$$

$$2. -\frac{1}{2} \int \frac{-2x}{\sqrt{1-x^2}} dx$$

$$= -\frac{1}{2} \left[2(1-x^2)^{1/2} \right] + C$$

$$3. \int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$$

$$6. \int \frac{f'(x)}{\sqrt{1-[f(x)]^2}} dx = \arcsin[f(x)] + C$$

$$\int 2^x dx = \frac{1}{\ln 2} 2^x + C$$

$$\frac{d}{dx} 2^x = 2^x \cdot \ln 2$$

$$\int 2^{f(x)} f'(x) dx = \frac{2^{f(x)}}{\ln 2} + C$$

$$7. \int a^{f(x)} f'(x) dx = \frac{a^{f(x)}}{\ln a} + C$$

$$1. \frac{1}{3} \int \frac{1}{3+3x} dx = \frac{1}{3} \ln |3+3x| + C$$

$$3. \int \frac{x}{1+(x)^2} dx = \int \frac{1+x}{x} dx = \int \frac{1}{x} + \frac{x}{x} dx$$

$$2. \int \frac{1}{3+x^2} dx = \frac{1}{3} \int \frac{1}{1+\frac{x^2}{3}} dx = \frac{\sqrt{3}}{3} \int \frac{\frac{1}{\sqrt{3}}}{1+(\frac{x}{\sqrt{3}})^2} dx = \frac{\sqrt{3}}{3} \arctan \frac{x}{\sqrt{3}} + C$$

$$2. \int \frac{1}{3+x^2} dx = \int \frac{1}{(\sqrt{3})^2 + (x)^2} dx = \frac{1}{\sqrt{3}} \arctan \frac{x}{\sqrt{3}} + C$$

$$= \ln|x| + x + c$$

$$3. \int \frac{\sqrt{x+1}-1}{\boxed{1+x}} dx$$

$$= \int \frac{\cancel{1+x}-1}{\cancel{1+x}} - \frac{1}{1+x} dx$$

$$= x - \ln|1+x| + c$$

List of Standard Integrals

$$1. \int [f(x)]^n f'(x) dx = \frac{f(x)^{n+1}}{n+1} + C$$

$$2. \int e^{f(x)} \cdot f'(x) dx = e^{f(x)} + C$$

$$3. \int a^{f(x)} \cdot f'(x) dx = \frac{a^{f(x)}}{\ln a} + C, \quad a > 0$$

$$4. \int \sin f(x) \cdot f'(x) dx = -\cos f(x) + C$$

$$5. \int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C$$

$$6. \int \frac{f'(x)}{1 + [f(x)]^2} dx = \tan^{-1}(f(x)) + C$$

$$7. \int \frac{f'(x)}{\sqrt{1 - [f(x)]^2}} dx = \sin^{-1}(f(x)) + C$$

$$6b \int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

$$7b \int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \frac{x}{a} + C$$

Evaluate the following integrals

$$1. \int \frac{-5x^4}{1-x^5} dx = -\frac{1}{5} \ln|1-x^5| + C$$

$$2. \int \frac{x^4}{1+x^{10}} dx = \frac{1}{5} \int \frac{5x^4}{1+(x^5)^2} = \frac{1}{5} \arctan x^5 + C$$

$$\begin{aligned} 3. \int \frac{1+x^5}{x^4} dx &= \int \left(\frac{1}{x^4} + x \right) dx \\ &= \int x^{-4} + x dx \\ &= -\frac{1}{3} x^{-3} + \frac{1}{2} x^2 + C \end{aligned}$$

$$4. \int \frac{\sec^2 x}{\sqrt{4 + \tan x}} dx = \int (4 + \tan x)^{-1/2} \sec^2 x dx$$

$$= 2(4 + \tan x)^{1/2} + C$$

$$5. \int \frac{e^{\sqrt{x}}}{2\sqrt{x}} dx$$

$$= 2 \cdot e^{\sqrt{x}} + C$$

$$6. -\frac{1}{6} \int \frac{-6x^5}{\sqrt{16 - x^6}} dx = -\frac{1}{6} \left[2(16 - x^6)^{1/2} \right] + C$$

$$7. \int \frac{3^x}{\sqrt{1 - 3^x}} dx$$

$$= \frac{1}{\ln 3} \int 3^x (1 - 3^x)^{-1/2} dx$$

$$= -\frac{1}{\ln 3} \cdot 2(1 - 3^x)^{1/2} + C$$

$$8. \int \frac{e^{3x}}{\sqrt{1 - e^{6x}}} dx$$

$$= \frac{1}{3} \int \frac{e^{3x}}{\sqrt{1 - (e^{3x})^2}} dx$$

$$= \frac{1}{3} \arcsin(e^{3x}) + C$$

$$\begin{aligned}
 9. \int \frac{x^2}{\sqrt{16-x^6}} dx &= \int \frac{x^2}{4\sqrt{1-\frac{x^6}{16}}} dx \\
 &= \frac{1}{3} \int \frac{\frac{3}{4} x^2}{\sqrt{1-\left(\frac{x^3}{4}\right)^2}} dx \\
 &= \frac{4}{3} \cdot \frac{1}{4} \arcsin \frac{x^3}{4} + C \\
 &= \frac{1}{3} \arcsin \frac{x^3}{4} + C
 \end{aligned}$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \arcsin \frac{f(x)}{a} + C$$

$$9. \frac{1}{3} \int \frac{\frac{3}{4} x^2}{\sqrt{(4)^2 - (x^3)^2}} dx = \frac{1}{3} \arcsin \frac{x^3}{4} + C$$

$$\begin{aligned}
 10. \int \frac{\sin 6x}{(1 + \cos 6x)^3} dx &= -\frac{1}{6} \int -6 \sin 6x (1 + \cos 6x)^{-3} dx \\
 &= -\frac{1}{6} \left[\frac{(1 + \cos 6x)^{-2}}{-2} \right] + C \\
 &= +\frac{1}{12} (1 + \cos 6x)^{-2} + C
 \end{aligned}$$

$$\begin{aligned}
 11. -\frac{1}{5} \int 8 \operatorname{cosec} 5x \cot 5x dx \\
 = -\frac{1}{5} \left[\frac{8 \operatorname{cosec} 5x}{\ln 8} \right] + C
 \end{aligned}$$

$$12. \int \tan 2x \, dx$$

$$= -\frac{1}{2} \int \frac{\sin 2x}{\cos 2x} \, dx = -\frac{1}{2} \ln |\cos 2x| + C$$

$$\frac{d}{dx} \frac{1}{x} = -\frac{1}{x^2}$$

$$13. \int \frac{\sinh(\frac{1}{x})}{-x^2} \, dx$$

$$= -\cosh(\frac{1}{x}) + C$$

$$14. \frac{1}{2} \int \frac{2x \cosh(x^2)}{3} \cdot 3 \frac{\sinh(x^2)}{\ln 3} \, dx$$

$$= \frac{1}{2} \cdot 3 \frac{\sinh(x^2)}{\ln 3} + C$$

$$15. \int \frac{\operatorname{cosec}^2(\ln x)}{x} \, dx$$

$$= -\cot(\ln x) + C$$

$$16. \frac{1}{3} \int 3x^2 e^{x^3} \sec(e^{x^3}) \tan(e^{x^3}) \, dx$$

$$= \frac{1}{3} \sec(e^{x^3}) + C$$

OR

$$\frac{1}{3} \int 3x^2 e^{x^3} \sec^2(e^{x^3}) (\tan(e^{x^3}))^1 \, dx$$

$$= \frac{1}{3} \left[\frac{\tan(e^{x^3})}{2} \right]^2 + C$$

$$17. \int (1-x^3)^2 dx = \int (1-2x^3+x^6) dx$$

$$= x - \frac{1}{2}x^4 + \frac{1}{7}x^7 + C$$

$$18. \int \frac{1}{4+x^2} dx = \int \frac{1}{(\frac{x}{2})^2 + (\frac{x}{2})^2} dx$$

$$= \frac{1}{2} \arctan \frac{x}{2} + C$$

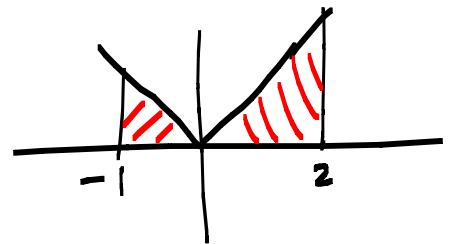
$$\boxed{\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \arctan \frac{f(x)}{a} + C}$$

$$19. \int_{-1}^2 |x| dx$$

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

$$= \int_{-1}^0 -x dx + \int_0^2 x dx$$

$$\quad \quad \quad -\frac{1}{2}x^2 \quad \quad \quad \frac{1}{2}x^2$$



$$20. \int (\sin x) \cos x dx$$

$$= \frac{1}{2} \sin^2 x + C \quad \underline{\text{or}}$$

$$-\frac{1}{2} (\cos x)^2 + C$$